ClinicaAI

Methodology, comparative evaluation and evidence limitations |

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Purpose: Explain the reasoning behind the generative virtual patient, the proposed comparison, what has actually been observed, and what remains to be verified.

1 Problem and early user insight

Students can recall theoretical knowledge but still struggle to solve clinical cases. The creator identified this need through personal experience and informal peer conversations.

Same 30 students · Musculoskeletal module	Above 10/20
Theory examination	83%
Clinical-case assessment	32%
Difference	51 percentage points

These percentages are descriptive, supplied by the project creator, and not independently audited. They do not establish that practice opportunities caused the gap or that ClinicaAI improves outcomes.

2 Why a generative virtual patient?

A preliminary small-sample exploration led the project to favour generative AI for flexible, contextual patient dialogue. The initial sample was too small to conclude that it is the best method. The AI Comparison Lab was created to make that choice testable rather than assume it.

3 Three approaches under one clinical reference

Rules-only: Deterministic matching against predefined case facts; consistent but limited flexibility.

Semantic matching: Finds semantically similar case information; may handle paraphrases but can return poorly matched responses.

Generative AI: Uses Google Gemini API to generate contextual answers; potentially more flexible, with risks of invented or contradictory facts.

For a fairer comparison, apply the same case dossier and questions to all approaches, log answers before revealing engine identity, and avoid disclosing the blind key to physician reviewers during scoring.

4 Benchmark and proposed study protocol

Development set: Q01–Q50, used for identifying failures and refining methods. **Evaluation set:** Q51–Q100, held out from tuning and used for the comparison. This partition must be respected in practice; simply labeling it 50/50 does not guarantee independence.

Planned outcome measures: question understanding, clinical accuracy/fidelity, conversational naturalness, invented or contradictory content, and important safety omissions. Report distributions and disagreement rates; do not present an unvalidated average score as a clinical safety certificate.

5 Human oversight and current evidence

ClinicaAI includes a blinded physician-review workflow: reviewers can import de-identified A/B/C responses, assign ratings and comments, check a clinical dossier, and export assessments. Two physicians provided encouraging initial feedback, but a completed, independently documented review of all 100 responses by each physician has not been established.

Illustrative student case: a fictional postoperative dyspnoea attempt received an indicative AI score of **55/100**. The feedback flagged an inappropriate option involving routine thrombolysis in a stable patient. This illustrates a feature of the prototype; it is not a prospective clinical safety validation.

6 Limitations and corrective actions

• Limited 30-student sample and unknown comparability of theory and clinical-case exam difficulty.

• One benchmark scenario does not establish generalizability across medical specialties or patient groups.

• Generative answers may hallucinate; scoring may undervalue sound free-text reasoning when form fields are empty.

• Benchmark reference answers are not automatically independently medically verified.

• AI-simulated reviewer records must never be attributed to Doctor1/Doctor2 without their actual exported decisions.

• Reviewer data is browser-local and must be explicitly exported for reproducible analysis.

7 Next three months — focused plan

Month 1: broaden and validate case content; fix reproducible scoring and language issues. **Month 2:** conduct documented student usability testing on realistic tasks. **Month 3:** obtain independent blinded physician reviews, analyse disagreements and unsafe answers, and publish the limitations.

Research perspective

ClinicaAI is an educational prototype. Its value rests on a working learning experience, clear methodological boundaries, and transparent follow-up studies. Current observations remain exploratory; independently documented clinical review is the next priority.